

Second-Moment Wedge Projection and the FVGQ Risk-Sharing Channel

A Standalone Research Report

Caught in the Crossfire Research Notes

May 1, 2026

Abstract

This report consolidates four research notes written on May 1, 2026. The goal is to turn the informal argument that “a first-order BCA prototype projects second-moment uncertainty into first-order wedges” into a self-contained theoretical document. The central result is a wedge projection equation: when the true data-generating process contains stochastic volatility, the investment wedge recovered by a Business Cycle Accounting (BCA) prototype must satisfy a forward-looking equation forced by the second-order risk correction in the true Euler equation. If the stochastic-volatility state is approximately AR(1), the projection admits a closed-form solution. Applied to a Fernandez-Villaverde–Guerron-Quintana (FVGQ) style collateral model, the second-order term can be decomposed into precautionary, risk-premium, collateral-risk, and two-agent aggregation components. The report then contrasts this second-order collateral channel with a first-order working-capital channel, which projects into the after-tax labor wedge rather than the investment wedge.

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1 Purpose and Main Results

This report integrates the following four internal research notes:

- 2026-05-01-wedge-projection-derivation-skeleton.md
- 2026-05-01-fvgq-Phi-closed-form.md
- 2026-05-01-fvgq-Phi-numerical-validation.md
- 2026-05-01-fvgq-two-agent-risk-sharing-wedge.md

The common question is how a first-order BCA prototype absorbs the second-order uncertainty terms generated by a nonlinear DSGE model with stochastic volatility.

The main conclusions are as follows.

1. The BCA investment wedge satisfies the projection equation

$$\hat{\tau}_{x,t} = \omega_\tau \mathbb{E}_t \hat{\tau}_{x,t+1} + \frac{1}{2} \Phi(\Omega_t),$$

where $\Phi(\Omega_t)$ is the second-order risk correction in the true Euler equation.

2. If Ω_t follows an AR(1) process, the projection has the closed form

$$\hat{\tau}_{x,t} = \frac{\Phi(\bar{\Omega})}{2(1 - \omega_\tau)} + \frac{\phi}{2(1 - \omega_\tau \rho_\Omega)} (\Omega_t - \bar{\Omega}), \quad \phi = \Phi'(\bar{\Omega}).$$

Thus BCA can transform a second-moment stochastic-volatility signal into a persistent first-order investment wedge.

3. In an FVGQ-style financial-friction model, the risk correction can be decomposed as

$$\Phi^{\text{FVGQ}} = \Phi^{\text{prec}} + \Phi^{\text{rp}} + \Phi^{\text{coll}} + \Phi^{\text{agg}}.$$

These terms capture precautionary motives, risk premia, collateral risk, and the aggregation error induced by using total consumption in a two-agent economy.

4. Financial uncertainty in the FVGQ collateral mechanism is predicted to appear mainly in the investment wedge, with a positive sign under the baseline collateral-risk channel. The sign of macro uncertainty is not guaranteed, because the risk-premium covariance can dominate the precautionary component.
5. A working-capital mechanism has a different empirical signature. If firms must finance wage payments in advance, the BCA after-tax labor wedge satisfies

$$\hat{\Lambda}_{\ell,t} = -\lambda_{wc}(q_t - \bar{q}) + O((q_t - \bar{q})^2), \quad \lambda_{wc} > 0.$$

Therefore a rise in the external finance spread lowers the after-tax labor wedge. This sign is first-order and does not depend on Hessian terms.

2 Notation and Normalization

Hats denote log deviations from the deterministic steady state; for example, $\hat{C}_t = \log(C_t/\bar{C})$. The stochastic-volatility state is

$$\Omega_t = (\sigma_{A,t}^2, \sigma_{e,t}^2)', \quad \Delta\Omega_t = \Omega_t - \bar{\Omega}.$$

In the one-factor case, Ω_t can be treated as a scalar.

To avoid ambiguity in the labor-wedge sign convention, this report uses the after-tax labor wedge

$$\Lambda_{\ell,t} \equiv 1 - \tau_{\ell,t}^{\text{tax}}.$$

Thus $\hat{\Lambda}_{\ell,t} < 0$ means that the labor wedge deteriorates. If the same result is reported in terms of the labor-tax rate $\tau_{\ell,t}^{\text{tax}}$, the sign is reversed.

Remark 1 (Normalization of Φ). This report uses the convention

$$\hat{\tau}_{x,t} = \omega_\tau \mathbb{E}_t \hat{\tau}_{x,t+1} + \frac{1}{2} \Phi(\Omega_t).$$

Some of the underlying notes absorb the factor $1/2$ into the definition of Φ . All formulas for ϕ_A , ϕ_e , $\kappa_{x,A}$, and $\kappa_{x,e}$ in this report use the convention above. A different convention rescales the coefficients but leaves the sign-identification logic unchanged.

3 Generic BCA Wedge Projection

3.1 BCA Prototype Euler Equation

The representative-household Euler equation in a CKM-style BCA prototype can be written, after first-order log linearization around the deterministic steady state, as

$$\hat{\tau}_{x,t} - \gamma \tilde{C}_t = \mathbb{E}_t \left[-\gamma \tilde{C}_{t+1} + \omega_K \tilde{R}_{t+1}^k + \omega_\tau \hat{\tau}_{x,t+1} \right]. \quad (\text{BCA-E})$$

Here γ is relative risk aversion, $\tilde{C}_t$ and $\tilde{R}_{t+1}^k$ are the observed consumption and capital-return objects fed into the BCA prototype, and ω_K, ω_τ are steady-state coefficients.

Suppose that the true data-generating process contains stochastic volatility. Its Euler equation, expanded to second order, can be written as

$$-\gamma \tilde{C}_t = \mathbb{E}_t \left[-\gamma \tilde{C}_{t+1} + \omega_K \tilde{R}_{t+1}^k \right] + \frac{1}{2} \Phi(\Omega_t) + o(\|\Delta\Omega_t\|). \quad (\text{DGP-E})$$

The term $\Phi(\Omega_t)$ collects risk corrections generated by conditional variances, conditional covariances, and Hessian terms.

Feeding the data generated by (DGP-E) into the BCA prototype and comparing (BCA-E) with (DGP-E) gives the projection equation

$$\boxed{\hat{\tau}_{x,t} = \omega_\tau \mathbb{E}_t \hat{\tau}_{x,t+1} + \frac{1}{2} \Phi(\Omega_t)}. \quad (\text{P})$$

This equation is the key object of the report.

3.2 Closed-Form Projection

Assumption 1 (Stochastic-volatility state). The stochastic-volatility state satisfies

$$\Delta\Omega_{t+1} = R_\Omega \Delta\Omega_t + \eta_{t+1}, \quad \rho(R_\Omega) < 1,$$

where $\rho(R_\Omega)$ is the spectral radius.

Assumption 2 (Local differentiability). Φ is differentiable near $\bar{\Omega}$ and satisfies

$$\Phi(\Omega_t) = \Phi(\bar{\Omega}) + \phi' \Delta\Omega_t + o(\|\Delta\Omega_t\|).$$

Theorem 1 (Wedge Projection). *Under the assumptions above and $|\omega_\tau| < 1$, the locally bounded solution to the projection equation (P) is*

$$\hat{\tau}_{x,t} = \frac{\Phi(\bar{\Omega})}{2(1-\omega_\tau)} + \frac{1}{2} \phi' (I - \omega_\tau R_\Omega)^{-1} \Delta\Omega_t + o(\|\Delta\Omega_t\|). \quad (\text{T1})$$

If Ω_t is a scalar AR(1) process, $\Delta\Omega_{t+1} = \rho_\Omega \Delta\Omega_t + \eta_{t+1}$, then

$$\hat{\tau}_{x,t} = \frac{\Phi(\bar{\Omega})}{2(1-\omega_\tau)} + \underbrace{\frac{\phi}{2(1-\omega_\tau \rho_\Omega)}}_{\kappa_x} (\Omega_t - \bar{\Omega}) + o(|\Omega_t - \bar{\Omega}|). \quad (\text{T1s})$$

Proof. Forward iteration of (P) gives

$$\hat{\tau}_{x,t} = \frac{1}{2} \sum_{j=0}^{\infty} \omega_\tau^j \mathbb{E}_t \Phi(\Omega_{t+j}).$$

The local approximation implies

$$\mathbb{E}_t \Phi(\Omega_{t+j}) = \Phi(\bar{\Omega}) + \phi' R_\Omega^j \Delta\Omega_t + o(\|\Delta\Omega_t\|).$$

Substitution and the geometric-series identities

$$\sum_{j=0}^{\infty} \omega_\tau^j = \frac{1}{1-\omega_\tau}, \quad \sum_{j=0}^{\infty} (\omega_\tau R_\Omega)^j = (I - \omega_\tau R_\Omega)^{-1}$$

yield (T1). The scalar formula is the one-dimensional special case. $\square$

The theorem has three immediate implications.

- **Persistence pass-through.** BCA may recover a wedge that looks like a persistent AR(1) first-order wedge, even though the underlying force is a stochastic-volatility state.
- **Amplification.** If $\omega_\tau \rho_\Omega$ is close to one, the second-order risk correction is amplified by $(1 - \omega_\tau \rho_\Omega)^{-1}$.
- **Sign discipline.** The sign of the investment-wedge response is governed by $\phi = \Phi'(\bar{\Omega})$. This makes it possible to distinguish second-order Euler channels from first-order static-FOC channels.

4 FVGQ Closed Form for $\Phi(\Omega_t)$

4.1 Entrepreneur Euler Equation

Consider a simplified FVGQ environment with a representative entrepreneur, CRRA utility, an always-binding collateral constraint, and no investment adjustment costs. Let $\lambda_t = u'(c_t)$ be the entrepreneur's marginal utility of consumption. The bond and capital first-order conditions are

$$\lambda_t = \beta R_t \mathbb{E}_t \lambda_{t+1} + \mu_t, \quad (\text{F1})$$

$$\lambda_t = \beta \mathbb{E}_t [\lambda_{t+1} (R_{t+1}^k + 1 - \delta)] + \mu_t \theta_t (1 - \delta). \quad (\text{F2})$$

Eliminating the collateral multiplier μ_t gives

$$\lambda_t [1 - \theta_t (1 - \delta)] = \beta \mathbb{E}_t [\lambda_{t+1} (R_{t+1}^k + 1 - \delta)] - \beta R_t \theta_t (1 - \delta) \mathbb{E}_t \lambda_{t+1}. \quad (\text{E-Euler})$$

This Euler equation is the source of the FVGQ risk-correction terms.

4.2 Risk-Correction Decomposition

A second-order expansion of (E-Euler), using $\hat{\lambda}_{t+1} = -\gamma \hat{c}_{t+1}$ under CRRA preferences, organizes the stochastic-volatility terms as

$$\Phi^{\text{FVGQ}}(\Omega_t) = \Phi^{\text{prec}}(\Omega_t) + \Phi^{\text{rp}}(\Omega_t) + \Phi^{\text{coll}}(\Omega_t). \quad (\text{FVGQ-Phi})$$

The three components are

$$\Phi^{\text{prec}}(\Omega_t) = \frac{1}{2} \gamma^2 [1 - \bar{\theta} (1 - \delta)] \Sigma_{cc}(\Omega_t), \quad (\text{Prec})$$

$$\Phi^{\text{rp}}(\Omega_t) = -\gamma \beta (\bar{R}^k + 1 - \delta) \Sigma_{cR}(\Omega_t), \quad (\text{RP})$$

$$\Phi^{\text{coll}}(\Omega_t) = \beta \bar{R} \bar{\theta} (1 - \delta) \nu_\theta \sigma_{e,t}^2. \quad (\text{Coll})$$

Here

$$\Sigma_{cc}(\Omega_t) = \text{Var}_t(\hat{c}_{t+1}), \quad \Sigma_{cR}(\Omega_t) = \text{Cov}_t(\hat{c}_{t+1}, \hat{r}_{t+1}^k).$$

The interpretation is direct. The precautionary component is positive when uncertainty raises the conditional variance of future consumption. The risk-premium component is negative when consumption and capital returns are positively correlated. The collateral-risk component captures the effect of financial uncertainty on future borrowing capacity; if $\nu_\theta > 0$, this component raises the projected investment wedge.

4.3 Mapping to κ_x

Near the deterministic steady state, write

$$\Sigma_{cc}(\Omega_t) = a_{cc,A} \sigma_{A,t}^2 + a_{cc,e} \sigma_{e,t}^2, \quad (1)$$

$$\Sigma_{cR}(\Omega_t) = a_{cR,A} \sigma_{A,t}^2 + a_{cR,e} \sigma_{e,t}^2. \quad (2)$$

Then

$$\phi_A = \frac{1}{2} \gamma^2 [1 - \bar{\theta} (1 - \delta)] a_{cc,A} - \gamma \beta (\bar{R}^k + 1 - \delta) a_{cR,A}, \quad (\text{phiA})$$

$$\phi_e = \frac{1}{2} \gamma^2 [1 - \bar{\theta} (1 - \delta)] a_{cc,e} - \gamma \beta (\bar{R}^k + 1 - \delta) a_{cR,e} + \beta \bar{R} \bar{\theta} (1 - \delta) \nu_\theta. \quad (\text{phie})$$

Theorem 1 gives

$$\hat{\tau}_{x,t} = \kappa_{x,A} \Delta \sigma_{A,t}^2 + \kappa_{x,e} \Delta \sigma_{e,t}^2, \quad (\text{Kappa-map})$$

with

$$\kappa_{x,A} = \frac{\phi_A}{2(1 - \omega_\tau \rho_{\sigma_A})}, \quad \kappa_{x,e} = \frac{\phi_e}{2(1 - \omega_\tau \rho_{\sigma_e})}. \quad (\text{Kappa})$$

This mapping makes the FVGQ sign predictions testable. Financial uncertainty contains the collateral-risk term and is therefore expected to generate a positive investment-wedge projection in the baseline model. Macro uncertainty is ambiguous because the precautionary and risk-premium terms can have opposing signs.

5 Two-Agent Risk-Sharing Wedge

The full FVGQ model has workers and entrepreneurs. The BCA prototype, however, uses aggregate NIPA consumption C_t as the consumption of a representative agent. The marginal utility $u'(C_t)$ used by the BCA observer is therefore not the same as the entrepreneur's marginal utility $u'(c_{E,t})$, which is the object that prices investment in the true model. This difference creates an aggregation wedge.

5.1 Aggregation Identity

Let

$$C_t = \omega_W c_{W,t} + \omega_E c_{E,t}, \quad \omega_W = 0.94, \quad \omega_E = 0.06.$$

Define steady-state consumption shares

$$s_W = \frac{\omega_W \bar{c}_W}{\bar{C}}, \quad s_E = \frac{\omega_E \bar{c}_E}{\bar{C}}, \quad s_W + s_E = 1.$$

A second-order expansion of aggregate consumption gives

$$\hat{C}_t = s_W \hat{c}_{W,t} + s_E \hat{c}_{E,t} + \frac{1}{2} s_W s_E (\hat{c}_{E,t} - \hat{c}_{W,t})^2 + O(3). \quad (\text{Agg-C})$$

Let

$$\delta c_t \equiv \hat{c}_{E,t} - \hat{c}_{W,t}.$$

Under CRRA preferences,

$$\log u'(C_t) - \log u'(c_{E,t}) = -\gamma(\hat{C}_t - \hat{c}_{E,t}) = \gamma s_W \delta c_t - \frac{1}{2} \gamma s_W s_E \delta c_t^2 + O(3). \quad (\text{MU-gap})$$

The first term is a first-order risk-sharing gap between entrepreneurs and workers. The second term is a Jensen aggregation term.

5.2 Projection Contribution

Substituting the aggregation identity into the BCA Euler equation and subtracting the entrepreneur Euler equation yields

$$\hat{\tau}_{x,t} = \omega_\tau \mathbb{E}_t \hat{\tau}_{x,t+1} + \gamma s_W (\mathbb{E}_t \delta c_{t+1} - \delta c_t) + \frac{1}{2} \Phi^{\text{tot}}(\Omega_t). \quad (\text{P-2A})$$

The total risk correction is

$$\Phi^{\text{tot}} = \Phi^{\text{prec}} + \Phi^{\text{rp}} + \Phi^{\text{coll}} + \Phi^{\text{agg}}.$$

Under the normalization used in this report, the aggregation component is

$$\Phi^{\text{agg}}(\Omega_t) = \gamma_{SWE} \left[\gamma \text{Var}_t(\delta c_{t+1}) - \text{Cov}_t(\delta c_{t+1}, \hat{r}_{t+1}^k) \right]. \quad (\text{Agg-Phi})$$

If the factor 1/2 is instead absorbed into the definition of Φ , the right-hand side of (Agg-Phi) must be rescaled accordingly.

5.3 Linear Policy-Function Interpretation

Let the one-step-ahead innovation loading of consumption dispersion be

$$\delta c_{t+1} = \chi_A \varepsilon_{A,t+1} + \chi_\theta \varepsilon_{\theta,t+1} + \text{state terms},$$

and let the corresponding loading of capital returns be

$$\hat{r}_{t+1}^k = r_A \varepsilon_{A,t+1} + r_\theta \varepsilon_{\theta,t+1} + \text{state terms}.$$

If the two shocks are orthogonal, then

$$\text{Var}_t(\delta c_{t+1}) = \chi_A^2 \sigma_{A,t}^2 + \chi_\theta^2 \sigma_{e,t}^2, \quad (3)$$

$$\text{Cov}_t(\delta c_{t+1}, \hat{r}_{t+1}^k) = \chi_A r_A \sigma_{A,t}^2 + \chi_\theta r_\theta \sigma_{e,t}^2. \quad (4)$$

Hence the marginal aggregation coefficients are

$$\phi_A^{\text{agg}} = \gamma_{SWE} \chi_A (\gamma \chi_A - r_A), \quad (\text{aggA})$$

$$\phi_e^{\text{agg}} = \gamma_{SWE} \chi_\theta (\gamma \chi_\theta - r_\theta). \quad (\text{agge})$$

Financial uncertainty typically makes entrepreneur consumption much more sensitive to collateral shocks than worker consumption. Therefore $\text{Var}_t(\delta c_{t+1})$ should rise with $\sigma_{e,t}^2$, reinforcing the positive investment-wedge projection. Macro uncertainty is less clear because both workers and entrepreneurs are exposed to TFP shocks, and the covariance term can offset the variance term.

6 Working-Capital Counter-Hypothesis

The FVGQ baseline is a second-order Euler-equation channel. A working-capital friction is different: it is a first-order static labor-demand channel. It should therefore project into the labor wedge rather than the investment wedge.

6.1 Exact Labor-Wedge Projection

Suppose firms must finance a fraction θ_{wc} of the wage bill before production. Let q_t denote the external finance spread. The firm's static condition is

$$A_t F_{L,t} = w_t [1 + \theta_{wc} q_t]. \quad (\text{WC-FOC})$$

The BCA intratemporal labor condition is

$$-\frac{U_L(C_t, L_t)}{U_C(C_t, L_t)} = \Lambda_{\ell,t}^{BCA} A_t F_{L,t}. \quad (\text{BCA-L})$$

The true household condition gives $-U_L/U_C = w_t$. Combining these conditions gives the exact projection identity

$$\Lambda_{\ell,t}^{BCA} = \frac{w_t}{A_t F_{L,t}} = \frac{1}{1 + \theta_{wc} q_t}. \quad (\text{WC-P})$$

Log-linearizing around $\bar{q}$ yields

$$\hat{\Lambda}_{\ell,t}^{BCA} = - \underbrace{\frac{\theta_{wc}}{1 + \theta_{wc} \bar{q}}}_{\lambda_{wc} > 0} (q_t - \bar{q}) + O((q_t - \bar{q})^2). \quad (\text{WC-LIN})$$

If the spread satisfies the first-order approximation

$$q_t - \bar{q} = \alpha_\xi \hat{\xi}_t + \alpha_\Omega (\Omega_t - \bar{\Omega}) + o(\|\hat{\xi}_t, \Omega_t - \bar{\Omega}\|), \quad \alpha_\xi > 0, \quad \alpha_\Omega \geq 0, \quad (\text{RP-WC})$$

then

$$\hat{\Lambda}_{\ell,t}^{BCA} = -\lambda_{wc} \alpha_\xi \hat{\xi}_t - \lambda_{wc} \alpha_\Omega (\Omega_t - \bar{\Omega}) + O(\hat{\xi}_t^2) + O((\Omega_t - \bar{\Omega})^2). \quad (\text{T2})$$

6.2 Economic Content

If a financial tightening $\hat{\xi}_t > 0$ or higher uncertainty raises the external finance spread, then

$$\hat{\Lambda}_{\ell,t}^{BCA} < 0.$$

This sign does not depend on Hessian terms, risk-premium covariance, or the numerical size of a simulation. Moreover, if $\hat{\xi}_t$ is a first-moment shock, the labor-wedge response is $O(\sigma)$, while a pure stochastic-volatility Hessian projection into the investment wedge is $O(\sigma^2)$. The two mechanisms are therefore distinguishable by both wedge location and local order.

7 Unified Sign Identification

Theorem 2 (Wedge Sign Identification). *Consider two competing mechanisms.*

H_0^{coll} : **FVGQ-style second-order collateral channel.** *If the true DGP transmits financial frictions mainly through second-order risk corrections in the Euler equation, then*

$$\mathbb{E}_t[\hat{\tau}_{x,t} \mid \Delta \sigma_{e,t}^2] = \kappa_{x,e} \Delta \sigma_{e,t}^2 + o(\Delta \sigma_{e,t}^2),$$

where $\kappa_{x,e}$ is given by (Kappa). In the baseline FVGQ collateral mechanism, the collateral and aggregation components push $\kappa_{x,e}$ toward a positive sign. The after-tax labor wedge does not have a signed first-order negative term driven by the financing shock.

H_1^{wc} : **wage-bill working-capital channel.** *If the true DGP contains a wage-bill working-capital friction and uncertainty or financial tightening raises the external finance spread, then*

$$\mathbb{E}_t[\hat{\Lambda}_{\ell,t} \mid \Delta \Omega_t] = -\lambda_{wc} (\alpha_\xi \rho_{\xi\Omega} + \alpha_\Omega) \Delta \Omega_t + o(\Delta \Omega_t) < 0,$$

provided that $\alpha_\xi \rho_{\xi\Omega} + \alpha_\Omega > 0$. Working capital does not directly create a first-order $\alpha_\xi \hat{\xi}_t$ term in the capital Euler equation.

The testable signatures are summarized below.

Mechanism	Direct condition	Main BCA wedge	Sign pattern
FVGQ collateral / SV risk	Intertemporal Euler, second order	Investment wedge τ_x	$\kappa_{x,e} > 0$ in baseline; $\kappa_{x,A}$ parameter-dependent
Two-agent aggregation	BCA uses C , while investment uses c_E	Investment wedge τ_x plus dispersion term	Financial channel likely reinforces $\tau_x > 0$; macro channel ambiguous
Working capital	Static labor demand, first order	After-tax labor wedge Λ_ℓ	$\hat{\Lambda}_\ell < 0$ when financing spreads rise

Therefore, if the data or simulations show that financial uncertainty mainly lowers $\log(1 - \tau_\ell)$ while the investment-wedge signal is weak, the evidence supports the working-capital channel. If the signal is concentrated in the investment wedge, especially with a positive response of τ_x to financial uncertainty, the pattern is closer to the FVGQ collateral and aggregation channel.

8 Numerical Validation from the FVGQ Workspace

The numerical validation uses pruned third-order policy matrices from the FVGQ preliminary workspace. This section should be read as a diagnostic check of the closed-form derivation, not as a final calibration exercise.

8.1 Extracted Coefficients

The extracted coefficients are listed below. The consumption variable is `cspt`, which is composite per-capita consumption. It is not identical to entrepreneur consumption c_E .

Coefficient	Value
$a_{cc,A}$	4.844×10^{-1}
$a_{cc,e}$	4.115×10^{-4}
$a_{cR,A}$	1.886×10^{-2}
$a_{cR,e}$	-1.786×10^{-4}
ν_θ	0 at the first/second-order extraction

The strongest sanity check is the stochastic-volatility loading:

$$\frac{\text{ghxu}[c, \text{sigAt}, e_A]}{\text{ghu}[c, e_A]} = 0.7500 = \rho_{\sigma_A}.$$

This confirms that the channel by which the volatility state changes the one-step-ahead conditional variance is present in the Dynare policy function in exactly the way required by the closed-form derivation.

8.2 Canonical Calibration Used in the Validation

The validation uses the actual calibration in `fvgq2020_solveonly.mod`, not the earlier draft values $\rho_\sigma = 0.95$ and $\bar{\theta} = 0.20$.

Parameter	Value
γ	2.0
β	0.994
δ	0.01625
$\bar{\theta}$	0.0954
$\bar{R}^k$	0.0273
$\bar{R}$	$1/\beta \simeq 1.0060$
$\rho_{\sigma_A}, \rho_{\sigma_e}$	0.75
ω_τ	$\simeq 0.973$
$1 - \omega_\tau \rho_\sigma$	$\simeq 0.270$

The amplification factor is therefore about $1/0.270 \simeq 3.7$, not the roughly 15-fold amplification implied by the earlier, incorrect $\rho_\sigma = 0.95$ value.

8.3 Predicted κ_x and LP Comparison

Without the Monte Carlo correction for ν_θ and without the full two-agent aggregation correction, the extracted coefficients imply

Quantity	Value
ϕ_A	$+8.40 \times 10^{-1}$
ϕ_e	$+1.10 \times 10^{-3}$
$\kappa_{x,A}$	+1.554
$\kappa_{x,e}$	$+2.04 \times 10^{-3}$

Converted into the impact response to a one-unit volatility shock, these figures imply

$$u_A \rightarrow \hat{\tau}_x : +1.06 \times 10^{-4}, \quad u_e \rightarrow \hat{\tau}_x : +1.93 \times 10^{-7}.$$

The comparison with preliminary local-projection peaks is

Channel	Closed-form sign	LP peak sign	Magnitude note
$u_A \rightarrow \tau_x$	Positive	Negative at $h = 40$	Sign mismatch at the peak
$u_e \rightarrow \tau_x$	Positive	Positive at $h = 32$	Correct sign, too small

This mismatch should not be read as a failure of the projection theorem. There are three immediate reasons.

1. The closed form is an impact and local projection object, whereas the reported LP number is a long-horizon peak. The comparison should include $h = 0$, the zero crossing, and the peak.
2. ν_θ is zero in the first/second-order matrix extraction, but the collateral-risk effect is likely to require a pruned third-order Monte Carlo extraction.
3. **cspt** is composite consumption, not entrepreneur consumption. The two-agent aggregation wedge requires separate coefficients for c_E and c_W .

9 Research Implications and Next Steps

The derivation turns the Year 2 BCA methodology from an informal set of competing hypotheses into a theorem-based framework with testable wedge signatures. The next steps are:

1. **Fix the numerical comparison.** Extract the LP impulse response at $h = 0$, at the zero crossing, and at the peak, rather than comparing only the peak.
2. **Extract ν_θ by Monte Carlo.** Perturb $\sigma_{e,t}$ on a fixed grid and use pruned simulations to regress mean log marginal utility or the borrowing-capacity value component on the volatility state.
3. **Augment the Dynare outputs for two-agent dispersion.** Add $\delta c_t = \log c_{E,t} - \log c_{W,t}$ and $\hat{r}_t^k$ as reported objects, then extract $\chi_A, \chi_\theta, r_A, r_\theta$ and compute ϕ_A^{agg} and ϕ_e^{agg} .
4. **Implement the working-capital comparison model.** Add wage-bill financing to the FVGQ block and verify whether the sign theorem $\hat{\Lambda}_{\ell,t} < 0$ holds in simulation.
5. **Write the formal appendix.** Complete the second-order expansion of (E-Euler), the aggregation wedge derivation, and the working-capital labor projection, with a consistent Φ normalization throughout.

Remaining issues that do not undermine tractability include pruning corrections to the risk-adjusted steady state, identification with correlated multi-dimensional volatility states, the non-smooth Hessian created by occasionally binding collateral constraints, and the measurement gap between composite consumption and entrepreneur marginal utility.

A Algebraic Details

A.1 Forward Solution of the Projection Equation

Starting from

$$\hat{\tau}_{x,t} = \omega_\tau \mathbb{E}_t \hat{\tau}_{x,t+1} + \frac{1}{2} \Phi(\Omega_t),$$

repeated substitution gives, for any J ,

$$\hat{\tau}_{x,t} = \frac{1}{2} \sum_{j=0}^J \omega_\tau^j \mathbb{E}_t \Phi(\Omega_{t+j}) + \omega_\tau^{J+1} \mathbb{E}_t \hat{\tau}_{x,t+J+1}.$$

If the solution is locally bounded and $|\omega_\tau| < 1$, the last term converges to zero. Substituting $\mathbb{E}_t \Delta \Omega_{t+j} = R_\Omega^j \Delta \Omega_t$ yields Theorem 1.

A.2 Working-Capital Labor Wedge

The working-capital firm condition and household condition are

$$A_t F_{L,t} = w_t (1 + \theta_{wc} q_t), \quad -\frac{U_L}{U_C} = w_t.$$

The BCA labor condition requires a wedge such that

$$w_t = \Lambda_{\ell,t}^{BCA} A_t F_{L,t}.$$

Therefore

$$\Lambda_{\ell,t}^{BCA} = \frac{1}{1 + \theta_{wc} q_t}.$$

Taking logs and expanding around $\bar{q}$ gives

$$\log \Lambda_{\ell,t}^{BCA} = -\log(1 + \theta_{wc} q_t) \simeq -\frac{\theta_{wc}}{1 + \theta_{wc} \bar{q}}(q_t - \bar{q}).$$

This is equation (WC-LIN).

A.3 Two-Agent Consumption Aggregation

Let $c_{i,t} = \bar{c}_i \exp(\hat{c}_{i,t})$. Then

$$\frac{C_t}{\bar{C}} = s_W \exp(\hat{c}_{W,t}) + s_E \exp(\hat{c}_{E,t}).$$

Taking logs and expanding to second order gives

$$\hat{C}_t = s_W \hat{c}_{W,t} + s_E \hat{c}_{E,t} + \frac{1}{2} [s_W \hat{c}_{W,t}^2 + s_E \hat{c}_{E,t}^2 - (s_W \hat{c}_{W,t} + s_E \hat{c}_{E,t})^2] + O(3).$$

The bracketed term simplifies to $s_W s_E (\hat{c}_{E,t} - \hat{c}_{W,t})^2$, which proves (Agg-C).